

Client–Server using Named Pipes

Question

Write a C program to implement client-server communication using Named Pipes (FIFO). The client should send a message to the server and the server should send a response back.

You need **two** files:

server.c

client.c

```
        printf("Server: %s\n", response);
    }

    return 0;
}
saikrishna__90156@DESKTOP-2JF9H5T:~$ |
```

```
saikrishna__90156@DESKTOP-2JF9H5T:~$ nano Practical6.c
saikrishna__90156@DESKTOP-2JF9H5T:~$ gcc Practical6.c -o Practical6
saikrishna__90156@DESKTOP-2JF9H5T:~$ ./Practical6
Server started.
Waiting for client...
Client: hello sir
exit
^C
saikrishna__90156@DESKTOP-2JF9H5T:~$ cat Practical6B.c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <fcntl.h>
#include <unistd.h>

#define FIFO1 "client_to_server"
#define FIFO2 "server_to_client"

int main() {
    char message[200];
    char response[200];

    printf("Client started.\n");

    while (1) {
        printf("Enter message: ");

        fgets(message, sizeof(message), stdin);
        message[strcspn(message, "\n")] = '\0';

        int fd1 = open(FIFO1, O_WRONLY);
        write(fd1, message, strlen(message) + 1);
        close(fd1);

        if (strcmp(message, "exit") == 0) {
            break;
        }

        int fd2 = open(FIFO2, O_RDONLY);
```